Project Based

SPONSORSHIP PROPOSAL · JÜLICH SUPERCOMPUTING CENTRE (JSC)

Applied research challenges for students on the Lower Rhine

Project Based is a student-led non-profit initiative based in Kleve. We organise hackathons and challenge events in which interdisciplinary student teams address real problems posed by companies and research institutions in the region. We would be honoured to welcome the Jülich Supercomputing Centre as a founding partner of our first event.



OUR CONCEPT

Each event is built around a problem provided by a partner organisation, which we develop into a structured challenge. Interdisciplinary student teams work on it over one to two days under the supervision of the partner's experts. The partner thereby gains a range of perspectives and several approaches to the same question; the most successful teams remain in contact afterwards.

RELEVANCE FOR JSC

- The region around Kleve hosts faculties of engineering, computer science, data science and biology at Hochschule Rhein-Waal (HSRW), Radboud University and HAN University of Applied Sciences. These disciplines correspond closely to JSC's research areas.
- JSC already supports early-career talent through its student programmes and training courses. Project Based complements these efforts by engaging students at an earlier stage, before they select a thesis or master's topic.

POSSIBLE CHALLENGES IN YOUR RESEARCH AREAS

Fluids & solids engineering
Simulation of a real component, or development of its digital twin, with a proposal for an improved design.

Machine learning for physics
Training of a surrogate model that predicts simulation results in seconds rather than hours.

Biology
Modelling of protein dynamics and folding, with an analysis of the factors determining the structure.

Data-intensive science
Cleaning, exploration and visualisation of a large research dataset.

These examples are indicative only. The final challenge would be defined in consultation with your team.

REQUESTED SUPPORT

- **Event sponsorship**
  Funding for prizes, catering, venue costs and student travel.
- **Challenge brief**
  A research question from one of your laboratories, scoped for a one- to two-day event.
- **Mentoring and jury**
  Two or three JSC experts, on site in Kleve or via video conference.

BENEFITS FOR JSC

- Early contact with motivated students for student assistant positions, theses and the JSC guest student programme.
- Several teams addressing one of your research questions in parallel, each with an independent approach.
- Visibility as a founding partner, including your logo on our website, event materials and communications.
- Increased presence in the Lower Rhine region and in the neighbouring Dutch cities of Nijmegen and Arnhem.

PROPOSED MODEL FOR OUR FIRST EVENT

Format	Hackathon or challenge, one to two days
Location	Kleve
Timing	Planned for 2027; date to be agreed with partners
Jury	Experts from partner institutions

ACADEMIC SUPPORTERS

Faculty of Technology and Bionics, Hochschule Rhein-Waal

Prof. Dr. Achim Kehrein	Applied Mathematics
Dr. Thorsten Camps	Mathematics and Computer Science
Prof. Dr. Matthias Krauledat	Computer Science
Prof. Dr. Ronny Hartanto	Computer Engineering
Prof. Dr.-Ing. Stéphane Danjou	Mechanical and Plant Engineering, focus on Digital Product Development

THE TEAM

Ahmad Azzeh
Founder, Chief Officer
Mechanical Engineering

Pablo de Villota
Logistics
Mechatronic Systems Eng.

Antonio Quaranta
Logistics
Mechanical Engineering

Jakob-Theodor Feldmann
Technical
Engineering

Marco Brackelmanns
Legal and accounting
Taxation and Law

CONTACT

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We would welcome the opportunity to present the concept to you and your colleagues, in person or via video conference.